



TIME SERIES ANALYSIS

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Frequency-domain analysis of signals and systems



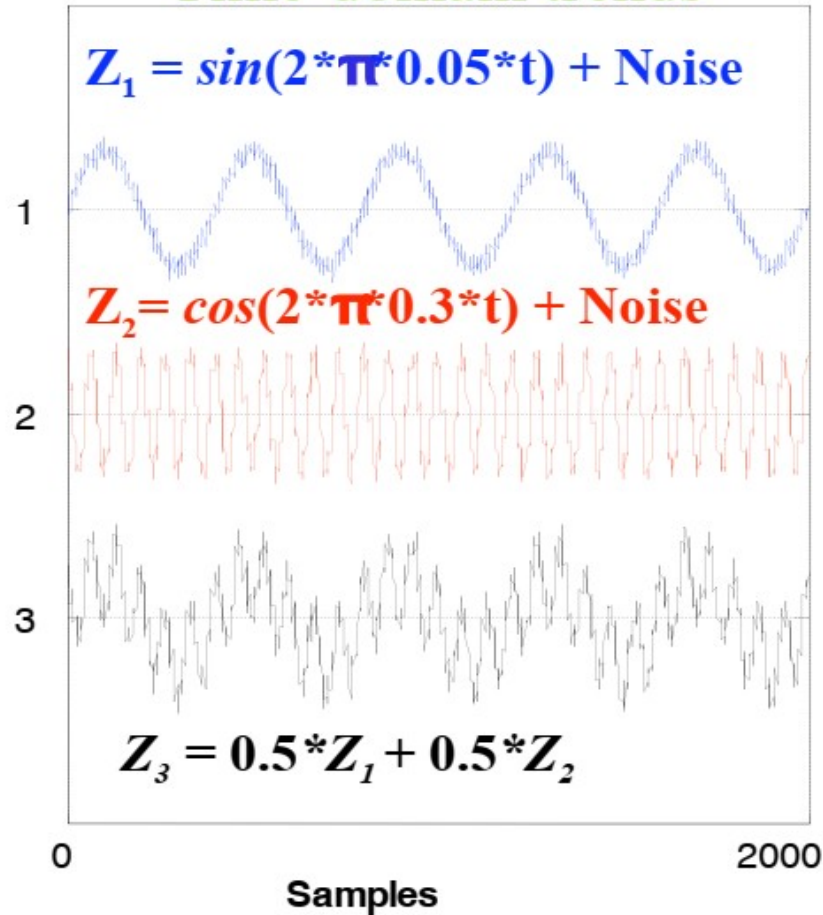
Power Spectral Density: Properties

- Extracts the power at each frequency or the “amount” of oscillations in that signal.
- Insensitive to time-delays/lags:
 - ▶ Spectrum of delayed or lagged signal is the same as that of the original signal (Delay is only reflected in the phase).
- Gives better estimates of the auto-correlation function (ACF):
- Outliers and small patches of missing values have negligible influence over the spectra.
 - ▶ Outliers are typically short-lived and therefore their spectra spread out uniformly across all frequencies
- Capable of capturing similarities in the oscillatory behaviour of signals that are shifted in time.
- Spectral analysis is a powerful tool in several areas such as acoustics, speech processing, chemometrics and process control

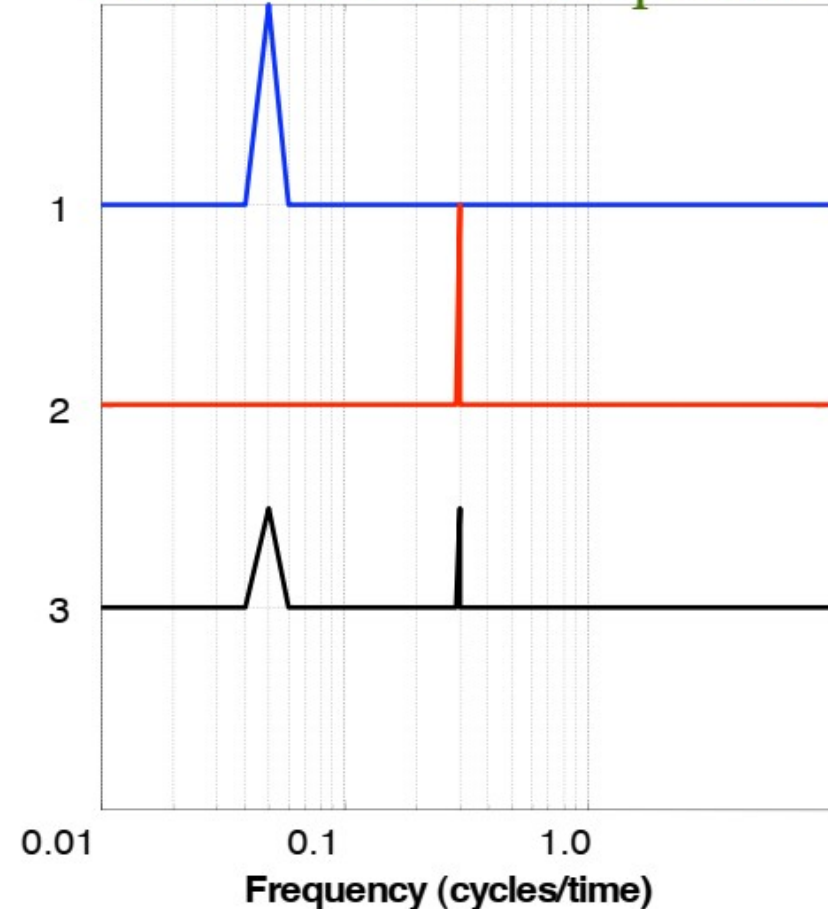
Identifying Linear Relationships

Discrete-Time Fourier Transform (DTFT)
(extracts the strength (power) of oscillations)

Time-domain trends



Normalized Power Spectra





Phase and Magnitude

- Magnitude or power spectrum as we have seen plays an important role in quantifying the “amount” or “strength” of each sinusoid in the signal
- However, what is also important, which is usually forgotten is to look for the time of occurrence of a particular frequency
 - ▶ When did a particular frequency contribute to the signal?
- This crucial information is contained in the **phase**
- The phase of the Fourier transform is not so easily interpretable due to computational reasons (as was seen above)
 - ▶ Nevertheless, some important qualitative judgments can be drawn without much ambiguity

Phase and Magnitude

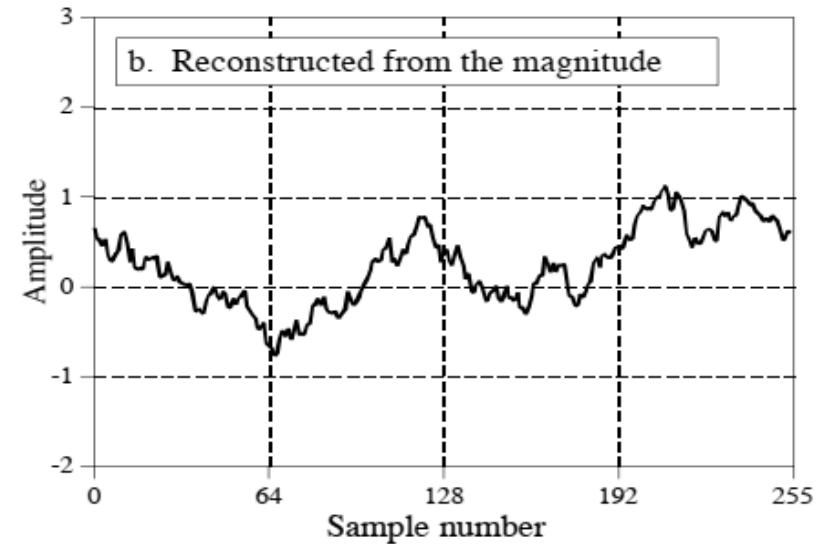
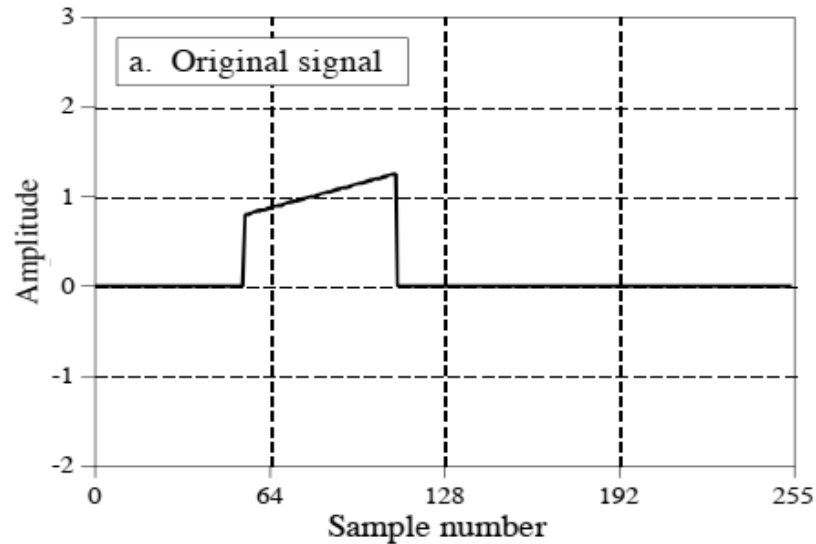
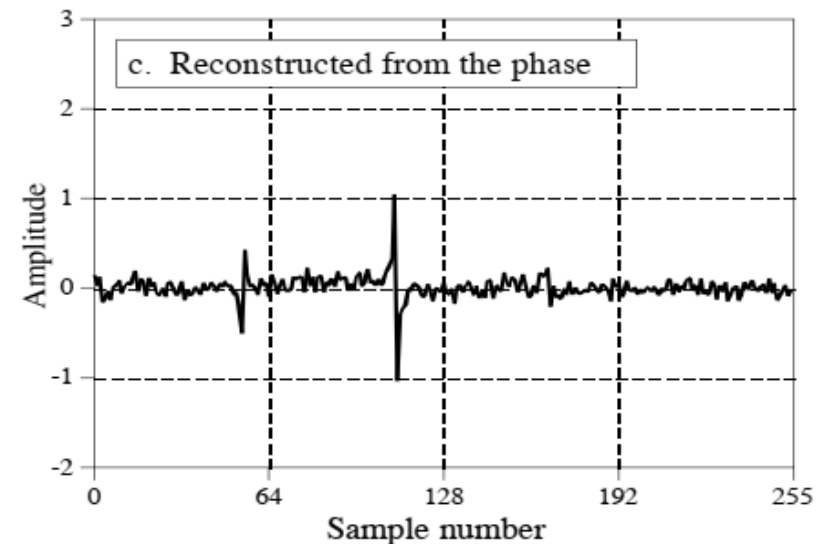


FIGURE 10-6

Information contained in the phase. Figure (a) shows a pulse-like waveform. The signal in (b) is created by taking the DFT of (a), replacing the *phase* with random numbers, and taking the Inverse DFT. The signal in (c) is found by taking the DFT of (a), replacing the *magnitude* with random numbers, and taking the Inverse DFT. The location of the *edges* is retained in (c), but not in (b). This shows that the phase contains information on the location of events in the time domain signal.





FT to LTI systems

Three important quantities arise with the application of Fourier Transform to *linear time-invariant systems*:

- ① **Frequency response function (FRF)**
- ② **Cross power spectrum**
- ③ **Coherence**

All the above quantities are meaningful only when the system satisfies both linearity and time-invariance

Each of these quantities is useful in several applications such as system identification, process control, design of filters and detection of non-linearities



Frequency Response Functions

FRF

The frequency response function of an LTI system $G(\omega)$ is defined as the Fourier Transform of the impulse response function $g[n]$ of the system

$$G(\omega) = \sum_{n=0}^{\infty} g[n]e^{-j\omega n} = \frac{Y(\omega)}{U(\omega)}$$

where $Y(\omega)$ and $U(\omega)$ are the Fourier Transforms of the output and input respectively

- The FRF is a **complex-valued** function of the *input frequency*
- $G(\omega) = |G(\omega)|e^{j\angle G(\omega)}$.
- When the input is a sinusoid $u(t) = A\sin(\omega t + \theta)$, the output is also a sinusoid of same frequency, but with a different amplitude and phase shift



Amplitude ratio and Phase Shift

- The output amplitude is given by $A|G(\omega)|$
- The phase of the output is given by $\theta + \angle G(\omega)$
- For general inputs, *i.e.*, when the inputs consist of mixed frequencies, the
 - ▶ The magnitude of FRF $|G(\omega)|$ provides the amplification / attenuation of the input amplitude at that frequency
 - ▶ The argument of FRF, $\phi(\omega) = \angle G(\omega)$ provides the phase shift of the input at that frequency
 - ▶ For e.g., if $u[n] = A_1 \sin(\omega_1 n) + A_2 \sin(\omega_2 n)$, the output is given by
$$y[n] = |G(\omega_1)|A_1 \sin(\omega_1 n + \angle G(\omega_1)) + |G(\omega_2)|A_2 \sin(\omega_2 n + \angle G(\omega_2))$$
- The combined graph of magnitude plot and phase plot is known as the **Bode plot**
 - ▶ The Bode plot is a very useful plot in linear systems analysis and filter design



Computation of Energy Density / Power Spectrum in Practice



Power Spectral Estimation

- *The theoretical spectrum is never known unless the analytical expression for the signal is known exactly*
- *In practice, one can only estimate the spectrum from the sampled sequence regardless of whether the signal is deterministic or random.*

In general, two methods exist for estimation of spectrum:

- 1 **Direct method:** Compute the Fourier transform of the signal and then take the squared magnitude (based on direct definition)
 - 2 **Autocorrelation method:** Compute the autocorrelation of the given sequence and then take its Fourier Transform (based on Wiener-Khintchine Theorem)
- Each of the above methods has its own applicability and limitations. It is very important to understand these facts since they determine the quality of the estimate as well as when they should be used.



Periodogram

In the frequency-domain literature, **periodogram** is one of the most commonly used terms to denote power spectrum of a signal

Periodogram

The periodogram obtained from N samples of a signal $x[n]$ is defined as

$$P_{xx}(f_k) = \frac{1}{N} \left| \sum_{n=0}^{N-1} x[n] e^{-j2\pi f_k n} \right|^2 = \frac{|X(f_k)|^2}{N} \quad (\text{Schuster, 1898})$$

- The periodogram is only an estimate of the “true” spectrum
- The actual spectrum is smoother and exhibits some statistical regularities
- The computed power at f_k (or ω_k) contains the power from the frequencies between the neighbouring bins (will be seen later)
- **By W-K Theorem, P_{xx} is also equal to the Fourier Transform of the ACF**



Deterministic Vs Stochastic

- **Deterministic signals:** The term *energy density spectrum* is applicable to finite duration deterministic sequences since they possess a finite energy.

Therefore, the *direct method* can be used for such signals in addition to the *indirect method*.

- **Random signals:** These signals, by virtue of random processes, exist infinitely in time and therefore possess infinite energy. However, they have finite average power and hence the term *power spectral density* is used for such signals.
 - ▶ Originally (before the advent of FFT), the autocorrelation method was used to estimate the power spectrum
 - ▶ Application of the direct method had two drawbacks (i) poor estimates and (ii) lack of a fast algorithm. Today, both have been overcome to a large extent by suitable smoothing and with the availability of FFT



Direct Method

- The Discrete Fourier Transform of the finite duration sequence is computed followed by the power spectrum computation
- We shall now observe an important point with regards to the **estimation** of the power spectrum of a finite-duration sequence
- Any finite-duration sequence can be treated as an infinite sequence viewed through a window of finite length. Mathematically, this may be written as $\tilde{x}[n] = x[n]w[n]$ where $\tilde{x}[n]$ is the length- L finite duration sequence, $x[n]$ is the infinite sequence and $w[n]$ is the **rectangular window** of length L

$$w[n] = \begin{cases} 1, & 0 \leq n \leq L - 1 \\ 0, & \text{otherwise} \end{cases}$$



Direct Method

- From the properties of FT we know that multiplication in time is equivalent to convolution in frequency domain:

$$\mathcal{F} \{ \tilde{x}[n] = x[n]w[n] \} \equiv \tilde{X}(\omega) = X(\omega) \star W(\omega) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\theta) W(\omega - \theta) d\theta$$

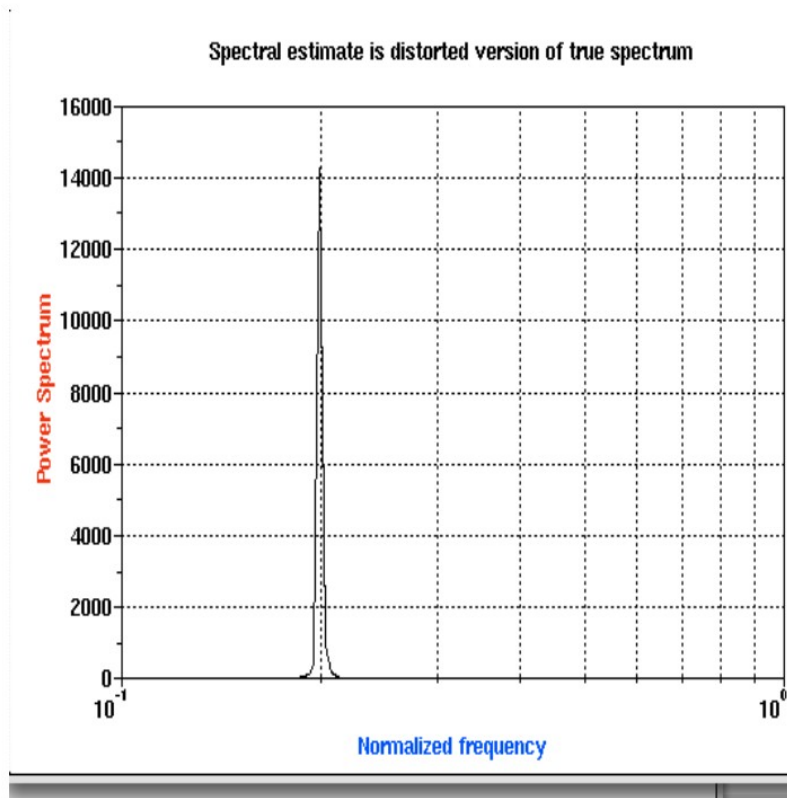
where $X(\cdot)$ and $W(\cdot)$ are the Fourier Transforms of the infinite sequence and the window functions respectively

- It is clear that what we obtain as an estimate is nothing but a convolution of the FT of the true signal with that of the rectangular window function!
- From above, it follows that the spectral estimate we obtain is a distorted version of the true spectrum
- In fact, linear systems theory tells us that convolution results in some form of distortion
 - ▶ For example, if a step signal is the input, the output is a distorted step-like signal

Example

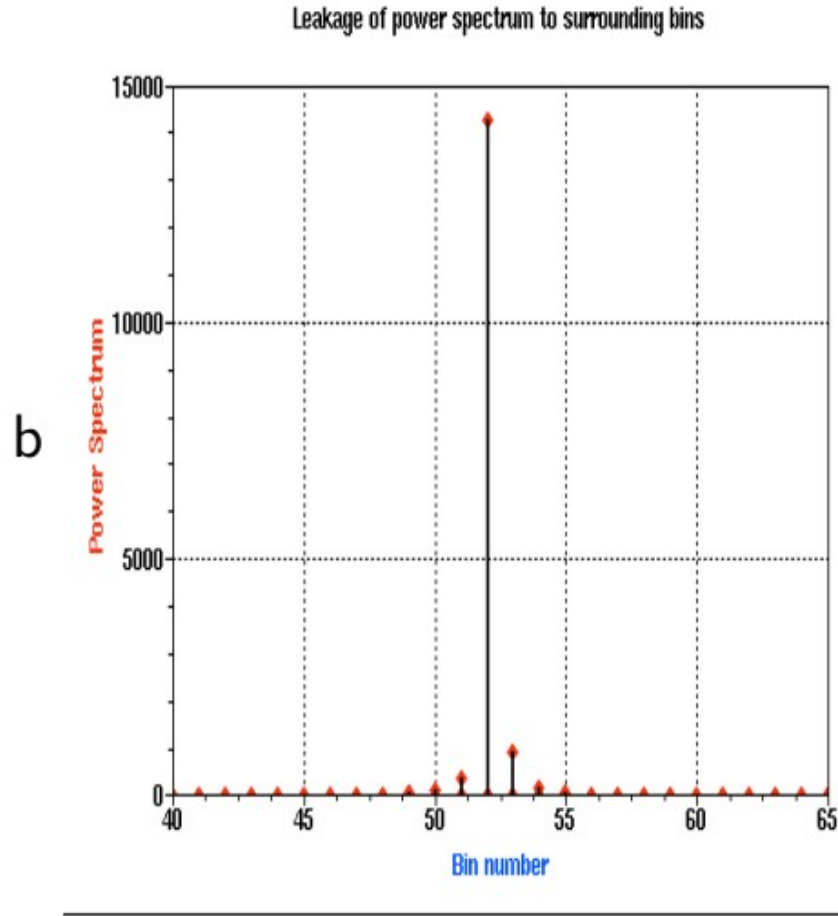
Suppose $x[n] = \cos(2\pi f_0 n)$ with $f_0 = 0.2$ cycles/sample and we have a finite data record of length $L = 256$. We are interested in estimating the spectrum of $x[n]$ using these set of samples

- We know that the true or theoretical spectrum is a delta function $\delta(f - f_0)$
- Observe the plot below



- The spectrum is not an ideal delta function as expected
- The estimate is in fact spread out over surrounding frequency bins
- This spread is termed as **spectral leakage**
- Spectral leakage can also occur due to mismatch between frequency sampling and the true frequency

Spectral Leakage



- The peak in the spectrum should have occurred at $f = 0.2$ cycles/sample
- The corresponding bin is $1 + \frac{0.2}{\Delta f} = 52.2$ where $\Delta f = 1/256$ is the frequency resolution. Now, there is no such bin as 52.2 (bins are integers)!
- The zero frequency corresponds to first bin
- Consequently, the power “leaks out” to bins 51 and 53
- The additional smearing that is observed is due to the *windowing*

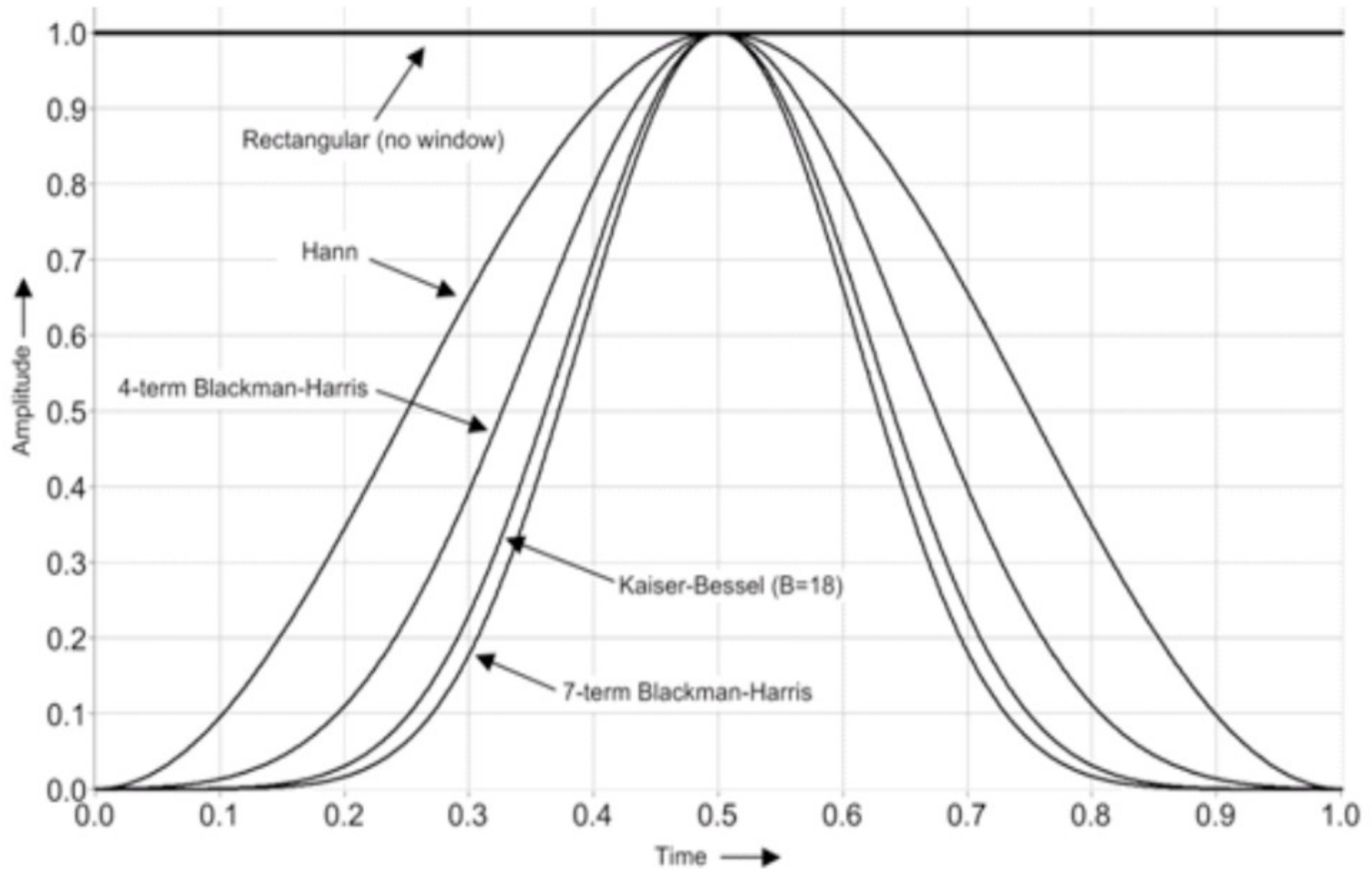
- In general, therefore, spectral leakage occurs due to:
 - 1 Windowing (convolution of the true spectrum)
 - 2 Lack of exactly matching basis function in the set of N basis functions $\{e^{-j2\pi k/N}\}$



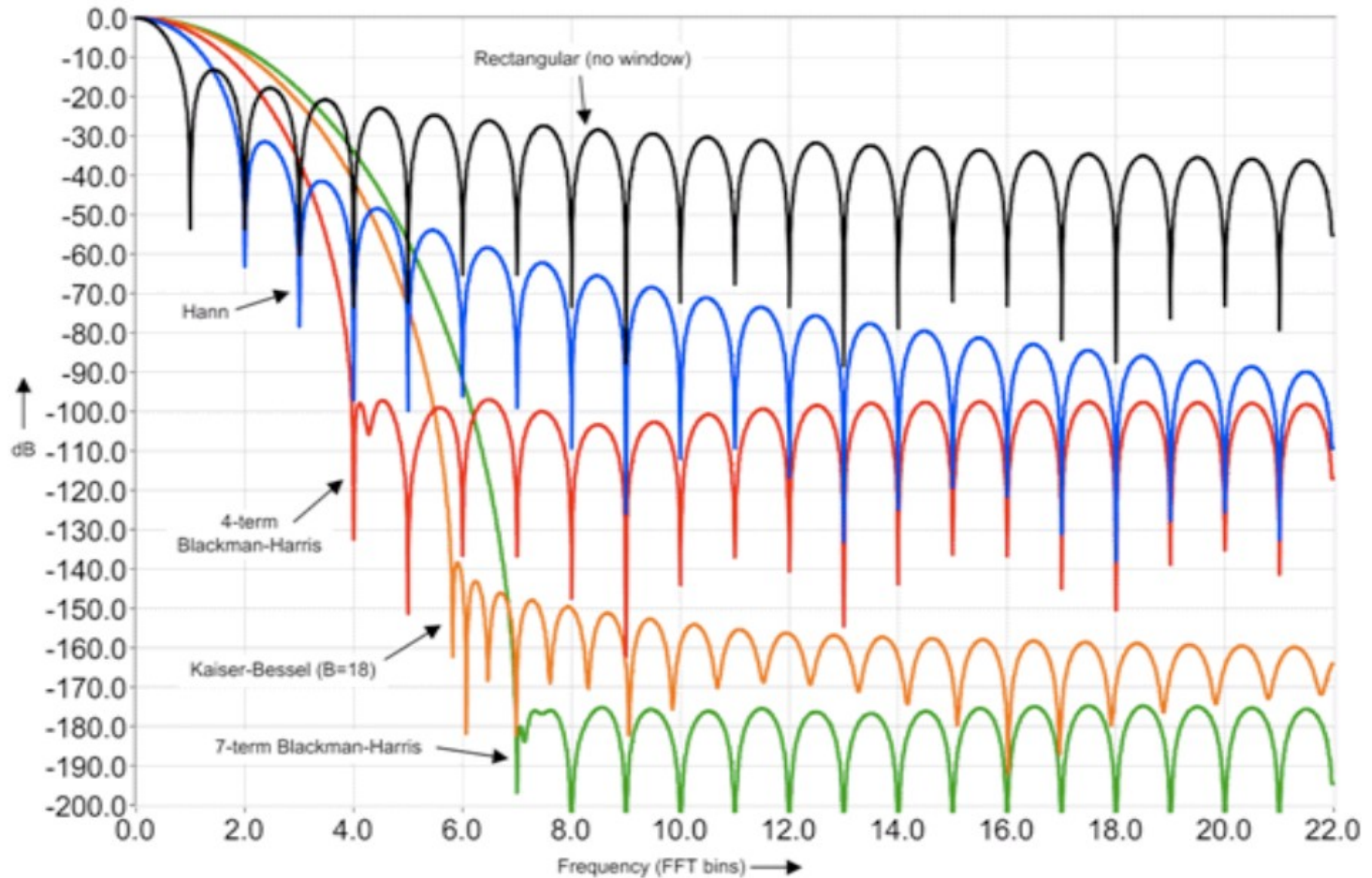
Overcoming Spectral Leakage

- One of the important consequences of the leakage is that two closely spaced frequencies cannot be *resolved*
- In general, spectral leakage can be changed by choosing different window functions
- Different windows are available with different trade-offs between spectral leakage and ability to resolve frequencies
 - ▶ The rectangular window has an excellent resolution but poor leakage characteristics
- Common windows $w[n]$ ($0 \leq n \leq N - 1$)
 - ▶ Rectangular: $w[n] = 1$
 - ▶ Hanning: $w[n] = 0.53836 - 0.46164 \cos\left(\frac{2\pi n}{N-1}\right)$
 - ▶ Hamming: $w[n] = 0.5 \left(1 - \cos\left(\frac{2\pi n}{N-1}\right)\right)$
 - ▶ Bartlett: $w[n] = \frac{2}{N} \left(\frac{N}{2} - \left|n - \frac{N-1}{2}\right|\right)$
 - ▶ Gaussian: $w[n] = e^{-\frac{1}{2} \left(\frac{n - (N-1)/2}{\sigma(N-1)/2}\right)^2}$ where $\sigma \leq 0.5$

Window Functions



Window Functions





Window Functions

Each window function has its own characteristics and suitability for different applications. The choice of the window function depends on the frequency content of the signal:

- If the signal contains strong interfering frequency components distant from the frequency of interest, choose a smoothing window with a high side lobe roll-off rate.
- If the signal contains strong interfering signals near the frequency of interest, choose a window function with a low maximum side lobe level.
- If the frequency of interest contains two or more signals very near to each other, spectral resolution is important. In this case, it is best to choose a smoothing window with a very narrow main lobe.



Window Functions

- If the amplitude accuracy of a single frequency component is more important than the exact location of the component in a given frequency bin, choose a window with a wide main lobe.
- If the signal spectrum is rather flat or broadband in frequency content, use the uniform window, or no window.
- In general, the Hanning window is satisfactory in 95% of cases. It has good frequency resolution and reduced spectral leakage.
- If the nature of the signal is not known, it is a good idea to start with the Hanning window.



Spectrum for Stochastic Signal

- Estimation of power spectrum or periodic components for time-series has been a subject of interest for more than 200 years now!
 - ▶ From Lagrange in 1770s through Fourier in early 1800s to the development of Hamming-Tukey, Blackman-Tukey methods and the most recent Multi-Tapering methods
- The estimation problem has to be treated differently from that of deterministic signals due to one fundamental reason
 - ▶ The usual way of estimating the power spectrum for deterministic signals yields a very poor estimate of the spectrum when applied to random signals
 - ▶ Of particular concern is the lack of decrease in the variance of the estimate even with the increase in the number of samples

$$\text{var}[P_{xx}(f)] = \Gamma_{xx}^2(f) \left[1 + \left(\frac{\sin(2\pi fN)}{N \sin(2\pi f)} \right)^2 \right] \implies \text{as } N \rightarrow \infty, \text{ var}[P_{xx}(f)] = \Gamma_{xx}^2(f)$$



Spectrum for Stochastic Signal

Method	Summary	Reference
Blackman-Tukey	Fourier transformation of smoothed, truncated autocovariance function	Chatfield, 1975
Smoothed periodogram	Estimate periodogram by DFT of time series; Smooth periodogram with modified Daniell filter	Bloomfield, 2000
Welch's method	Averaged periodograms of overlapped, windowed segments of a time series	Welch, 1967
Multi-taper method (MTM)	Use orthogonal windows (tapers) to get approximately independent estimates of spectrum; combine estimates	Percival and Walden, 1993
Singular spectrum analysis (SSA)	Eigenvector analysis of autocorrelation matrix to eliminate noise prior to transformation to spectral estimates	Vautard and Ghil, 1989
Maximum entropy (MEM)	Parametric method: estimate acf and solve for AR model parameters; AR model has theoretical	Kay, 1988



Spectrum for Stochastic Signal

Methods for estimating spectrum can be categorized into two classes:

1 Non-parametric methods:

- ▶ These are methods based on data records.
- ▶ ↑ No model is required to be fit to data.
- ▶ ↑ No major assumptions about the data sequence are made
 - ★ Two assumptions arise though: (1) Autocorrelation dies out after N and (2) Data is periodic with period N
- ▶ An important goal is to obtain a consistent estimate
- ▶ Reduction in variance is achieved at the cost of decrease in resolution
- ▶ ↓ Spectral leakage is an issue

2 Parametric methods:

- ▶ ↑ Eliminates the need for windows. Therefore, no leakage issues arise
- ▶ ↑ Provide better frequency resolution than FFT-based methods
- ▶ ↓ Requires a model-based description of the stochastic process
- ▶ Autocorrelation need not go to zero after N . The auto-correlation sequence can be computed from the model



Spectrum Via ACF

Blackman & Tukey (1958) developed a method to obtain the smoothed spectrum based on the Wiener-Khintchine Theorem.

B-T estimate

The estimate is given by:
$$\bar{P}_{xx}^{BT}(f) = \sum_{m=-(M-1)}^{M-1} r_{xx}[m]w[m]e^{-j2\pi fm}$$

where $r[.]$ is the estimated *auto-correlation* sequence and $w[.]$ is the *window* function used for smoothing the ACF.

- As noted earlier, the smoothing of the ACF is equivalent to convolution of the “raw” periodogram with the spectrum of the window function
- The window function should be symmetric (even) about $m = 0$ to ensure that the estimate of the power spectrum is real
- The window spectrum should be non-negative



Smoothing the Spectrum

1 Subtract mean and detrend time series

- ▶ *Removal of mean* is essential to remove the generally dominant DC component
- ▶ *Detrending* (removal of trends) is necessary to remove any seasonal / linear trends

2 Compute discrete Fourier transform (DFT)

- ▶ Remember to have a length equal to power of 2 for efficiency

3 Compute the (raw) periodogram

- ▶ Use $P_{xx}(f_k) = |DFT(x)|^2$
- ▶ You may verify that $\sum_{n=0}^{N-1} x^2[n] = \sum_{k=0}^{N-1} P_{xx}(f_k)$

4 Smooth the periodogram to obtain $\overline{P}_{xx}^D(f)$

- ▶ Use an appropriate window to smooth the spectrum



Smoothing the Spectrum

The Daniell window or filter (Bloomfield, 2000) is generally recommended for smoothing the spectrum (other alternatives are available)

The window of span (length) M is given by

$$\begin{aligned} g[m] &= \frac{1}{2(M-1)} \quad m = 1 \text{ or } M \\ &= \frac{1}{M-1} \quad \text{otherwise} \end{aligned}$$

The advantage of the above filter over the *rectangular* filter (window) is that it has less *spectral leakage*

- Smoothing is done by traversing this window (of suitable length) across the spectrum
 - ▶ Excessive smoothing can be detrimental to the spectral resolution while insufficient smoothing can result in erratic



Bartlett's Method

- 1 Divide the original sequence into K nonoverlapping segments of length M
- 2 Compute the periodogram estimate of each segment
 - ▶ Remove the means prior and trends prior to performing DFT

$$P_{xx}^{(i)}(f) = \frac{1}{M} \left| \sum_{n=0}^{M-1} x_i[n] e^{-j2\pi fn} \right|^2 \quad i = 1, \dots, K$$

- 1 Average the L periodograms to get $\bar{P}_{xx}^B(f)$

The Bartlett window is given by

$$w_B[m] = \begin{cases} 1 - \frac{|m|}{N}, & |m| < N \\ 0, & \text{otherwise} \end{cases}$$

and

the power spectrum is given by

$$\bar{P}_{xx}^B(f) = \frac{1}{K} \sum_{i=1}^K P_{xx}^{(i)}(f)$$

- The variance reduces by a factor L at the cost of reduction in frequency resolution by approximately L .



Welch's Method

Welch (1967) further modified Bartlett's method as follows

- ① Non-overlapping segments may now be overlapped
 - ▶ Overlapping is a tuning parameter (user's choice)
- ② Apply a temporal window, $w[n]$, to each segment before periodogram calculation
 - ▶ The choice of window could be *Hanning*, *Hamming*, *etc*

$$P_{xx}^{(i)}(f) = \frac{1}{MU} \left| \sum_{n=0}^{M-1} x_i[n]w[n]e^{-j2\pi fn} \right|^2 \quad i = 1, 2, \dots, L \quad \text{where } L \text{ is}$$

the number of segments, M is the number of samples in each segment and U is a normalization factor for the power in the window function

- ① Average the resulting periodograms to obtain $\bar{P}_{xx}^{WEL}(f)$

$$\bar{P}_{xx}^{WEL}(f) = \frac{1}{L} \sum_{i=1}^L P_{xx}^{(i)}(f)$$



Performance

It is useful to compare the performance of the above three methods, namely, B-T, Bartlett and Welch's methods. Define a performance measure (Proakis and Manolakis, 2005)

$$Q_A = \frac{\{E[P_{xx}^A(f)]\}^2}{\text{var}[P_{xx}^A(f)]}$$

- The numerator is the “true” (or expected) variance of the power spectral estimate while the denominator is the variance of the estimate
- The superscript A stands for the method under evaluation (A could be one of BT, B or W)
- The reciprocal of the above measure could also be used as a performance measure



Performance

- **Periodogram:** Raw estimate

$$\begin{aligned} E[P_{xx}(f)] &= \int_{-1/2}^{1/2} \Gamma_{xx}(\alpha) W(f - \alpha) d\alpha \longrightarrow \Gamma_{xx}(f) \\ \text{var}[P_{xx}(f)] &= \Gamma_{xx}^2(f) \\ Q_P &= \frac{\Gamma_{xx}^2(f)}{\Gamma_{xx}^2(f)} = 1 \end{aligned}$$

- Observe that the raw periodogram or the raw estimate is asymptotically unbiased
- However, Q_P is fixed and independent of the data length N . This is an indication of the poor quality of the estimate



Performance

- **Bartlett's method:** Segmenting the data into overlapping segments followed by smoothing the spectrum with triangular window

$$E[P_{xx}^B(f)] = \int_{-1/2}^{1/2} \Gamma_{xx}(\alpha) W_B(f - \alpha) d\alpha \longrightarrow \Gamma_{xx}(f) w_B(0) = \Gamma_{xx}(f)$$

$$\text{var}[P_{xx}^B(f)] = \frac{1}{K} \Gamma_{xx}^2(f)$$

$$Q_B = \frac{N}{M}$$

- Observe that here too the estimate is asymptotically unbiased
- The variance can be decreased by increasing K . Further, it goes to zero as $N \rightarrow \infty$ and therefore the estimate is consistent.
- However, the frequency resolution of the Bartlett estimate (by taking the 3-dB width of the main lobe of the rectangular window) is $\Delta f = \frac{0.9}{M}$. Therefore, if K is increased (for a fixed N) then the resolution goes down



Performance

- **Welch's method:** Modification of Bartlett's method by allowing overlapping and using a temporal window prior to DFT

$$E[P_{xx}^W(f)] = \int_{-1/2}^{1/2} \Gamma_{xx}(\alpha) W_W(f - \alpha) d\alpha \longrightarrow \Gamma_{xx}(f)$$

$$\text{var}[P_{xx}^W(f)] = \begin{cases} \frac{1}{L} \Gamma_{xx}^2(f), & \text{no overlap} \\ \frac{9}{8L} \Gamma_{xx}^2(f), & 50\% \text{ overlap} \end{cases}$$

$$Q_W = \begin{cases} L = \frac{N}{M}, & \text{no overlap} \\ \frac{8L}{9} = \frac{16N}{9M}, & 50\% \text{ overlap} \end{cases}$$

- Observe that here too the estimate is asymptotically unbiased
- As $N \rightarrow \infty$ and $M \rightarrow \infty$, variance tends to zero and therefore the estimate is consistent
- The frequency resolution of the Welch's estimate (for a triangular window) is $\Delta f = \frac{1.28}{M}$.
- From above $Q_B = 1.1N\Delta f$